

where q is the appropriately-weighted mean of the various $\mathbb{E}[H_{i\ell}H_{j\ell}]$ terms for which i, j, ℓ are not mutually distinct. Since $|H_{ij}| \leq 4\nu$, $\mathbb{E}[H_{i\ell}H_{j\ell}] < 16\nu^2$ and so $|q| \leq 16\nu^2$ as well. Noting that

$$\frac{n(n-1)(n-2)}{n^3} = 1 - \frac{3}{n} + \frac{2}{n^2}$$

we obtain

$$\left| \frac{1}{n^3} \sum_{ij\ell} \mathbb{E}[H_{i\ell}H_{j\ell}] - \mathbb{E}[H_{12}H_{13}] \right| = \left(\frac{3}{n} - \frac{2}{n^2} \right) |-\mathbb{E}[H_{12}H_{13}] + q| \leq \left(\frac{3}{n} - \frac{2}{n^2} \right) 32\nu^2. \quad (10)$$

The second term can be handled similarly:

$$\begin{aligned} \frac{1}{n^4} \sum_{ijab} \mathbb{E}[H_{ij}H_{ab}] &= \frac{1}{n^4} \sum_{ijab: \{i,j,a,b\}=4} \mathbb{E}[H_{ij}H_{ab}] + \frac{1}{n^4} \sum_{ijab: \{i,j,a,b\} < 4} \mathbb{E}[H_{ij}H_{ab}] \\ &= \frac{n(n-1)(n-2)(n-3)}{n^4} \mathbb{E}[H_{ij}H_{ab}] + \left(1 - \frac{n(n-1)(n-2)(n-3)}{n^4} \right) r, \end{aligned}$$

where r is the appropriately-weighted mean of the non-distinct terms, $|r| \leq 16\nu^2$. For i, j, a, b all distinct, $\mathbb{E}[H_{ij}H_{ab}] = \mathbb{E}[H_{12}]^2$. Here

$$\frac{n(n-1)(n-2)(n-3)}{n^4} = \frac{(n-1)(n^2-5n+6)}{n^3} = 1 - \frac{6}{n} + \frac{11}{n^2} - \frac{6}{n^3}$$

and so

$$\left| \frac{1}{n^4} \sum_{ijab} \mathbb{E}[H_{ij}H_{ab}] - \mathbb{E}[H_{12}]^2 \right| \leq \left(\frac{6}{n} - \frac{11}{n^2} + \frac{6}{n^3} \right) 32\nu^2. \quad (11)$$

Recalling $\sigma_k^2 = 4(\mathbb{E}[H_{12}H_{13}] - \mathbb{E}[H_{12}]^2)$,

$$|\mathbb{E} \hat{\sigma}_k^2 - \sigma_k^2| \leq 128\nu^2 \left(\frac{9}{n} - \frac{13}{n^2} + \frac{6}{n^3} \right),$$

and since $n \geq 1$, we have $13/n^2 > 6/n^3$, yielding the result. $\square$

Lemma 19. Under Assumptions (A) and (C), we have

$$\sup_{\omega, \omega' \in \Omega} \frac{|\hat{\sigma}_\omega^2 - \hat{\sigma}_{\omega'}^2|}{\|\omega - \omega'\|} \leq 256L_k\nu \quad \text{and} \quad \sup_{\omega, \omega' \in \Omega} \frac{|\sigma_\omega^2 - \sigma_{\omega'}^2|}{\|\omega - \omega'\|} \leq 256L_k\nu.$$

Proof. We first handle the change in $\hat{\sigma}_k$:

$$\begin{aligned} |\hat{\sigma}_{k\omega}^2 - \hat{\sigma}_{k\omega'}^2| &= 4 \left| \frac{1}{n^3} \sum_{ij\ell} H_{i\ell}^{(\omega)} H_{j\ell}^{(\omega)} - \frac{1}{n^3} \sum_{ij\ell} H_{i\ell}^{(\omega')} H_{j\ell}^{(\omega')} - \frac{1}{n^4} \sum_{ijab} H_{ij}^{(\omega)} H_{ab}^{(\omega)} + \frac{1}{n^4} \sum_{ijab} H_{ij}^{(\omega')} H_{ab}^{(\omega')} \right| \\ &\leq \frac{4}{n^3} \sum_{ij\ell} |H_{i\ell}^{(\omega)} H_{j\ell}^{(\omega)} - H_{i\ell}^{(\omega')} H_{j\ell}^{(\omega')}| + \frac{4}{n^4} \sum_{ijab} |H_{ij}^{(\omega)} H_{ab}^{(\omega)} - H_{ij}^{(\omega')} H_{ab}^{(\omega')}|. \end{aligned}$$

We can handle both terms by bounding

$$\begin{aligned} |H_{ij}^{(\omega)} H_{ab}^{(\omega)} - H_{ij}^{(\omega')} H_{ab}^{(\omega')}| &\leq |H_{ij}^{(\omega)} H_{ab}^{(\omega)} - H_{ij}^{(\omega)} H_{ab}^{(\omega')}| + |H_{ij}^{(\omega)} H_{ab}^{(\omega')} - H_{ij}^{(\omega')} H_{ab}^{(\omega')}| \\ &= |H_{ij}^{(\omega)}| |H_{ab}^{(\omega)} - H_{ab}^{(\omega')}| + |H_{ij}^{(\omega)} - H_{ij}^{(\omega')}| |H_{ab}^{(\omega')}| \\ &\leq 4\nu \left(|H_{ab}^{(\omega)} - H_{ab}^{(\omega')}| + |H_{ij}^{(\omega)} - H_{ij}^{(\omega')}| \right). \end{aligned}$$

Using Assumption (C) and the definition of H ,

$$|H_{ij}^{(\omega)} - H_{ij}^{(\omega')}| \leq 4L_k \|\omega - \omega'\|$$

so

$$|H_{ij}^{(\omega)} H_{ab}^{(\omega)} - H_{ij}^{(\omega')} H_{ab}^{(\omega')}| \leq 32\nu L_k \|\omega - \omega'\| \quad (12)$$

and hence

$$|\hat{\sigma}_\omega^2 - \hat{\sigma}_{\omega'}^2| \leq 256\nu L_k \|\omega - \omega'\|.$$

Again using (12), we also have

$$\begin{aligned} |\sigma_\omega^2 - \sigma_{\omega'}^2| &\leq 4|\mathbb{E}[H_{12}^{(\omega)} H_{13}^{(\omega)}] - \mathbb{E}[H_{12}^{(\omega')} H_{13}^{(\omega')}]| + 4|\mathbb{E}[H_{12}^{(\omega)}]^2 - \mathbb{E}[H_{12}^{(\omega')}]^2| \\ &\leq 4\mathbb{E}|H_{12}^{(\omega)} H_{13}^{(\omega)} - H_{12}^{(\omega')} H_{13}^{(\omega')}| + 4\mathbb{E}|H_{12}^{(\omega)} H_{34}^{(\omega)} - H_{12}^{(\omega')} H_{34}^{(\omega')}| \\ &\leq 256\nu L_k \|\omega - \omega'\|. \end{aligned} \quad \square$$

A.4. Constructing appropriate kernels

We now show Propositions 7 to 9, which each state that Assumption (C) is satisfied by various choices of kernel. The following assumption will be useful for different kernel schemes.

(I) The domain $\mathcal{X}$ is Euclidean and bounded, $\mathcal{X} \subseteq \{x \in \mathbb{R}^d : \|x\| \leq R_X\}$ for some constant $R_X < \infty$.

We begin by recalling a well-known property of the Gaussian kernel, useful for both Gaussian bandwidth selection and deep kernels. A proof is in Appendix A.5.

Lemma 20. *The Gaussian kernel $\kappa(a, b) = \exp\left(-\frac{\|a-b\|^2}{2\sigma^2}\right)$ satisfies*

$$|\kappa(a, b) - \kappa(a', b')| \leq \frac{1}{\sigma\sqrt{e}} (\|a - b\| + \|a' - b'\|) \leq \frac{1}{\sigma\sqrt{e}} (\|a - a'\| + \|b - b'\|).$$

A.4.1. GAUSSIAN BANDWIDTH SELECTION (PROPOSITION 7)

Lemma 20 immediately gives us Assumption (C) when we chose among Gaussian kernels:

Proposition 21. *Define a one-dimensional Banach space for inverse lengthscales of Gaussian kernels $\gamma > 0$, so that $k_\gamma(x, y) = \kappa_{1/\gamma}(x, y)$, with standard addition and multiplication and norms defined by the absolute value, and k_0 taken to be the constant 1 function. Let Ω be any subset of this space. Under Assumption (I), Assumption (C) holds: for any $x, y \in \mathcal{X}$ and $\gamma, \gamma' \in \Gamma$,*

$$|k_\gamma(x, y) - k_{\gamma'}(x, y)| \leq \frac{2R_X}{\sqrt{e}} |\gamma - \gamma'|.$$

Proof.

$$|k_\gamma(x, y) - k_{\gamma'}(x, y)| = |\kappa_1(\gamma x, \gamma y) - \kappa_1(\gamma' x, \gamma' y)| \leq \frac{1}{\sqrt{e}} |\gamma\|x - y\| - \gamma'\|x - y\|| = \frac{\|x - y\|}{\sqrt{e}} |\gamma - \gamma'|. \quad \square$$

A.4.2. DEEP KERNELS (PROPOSITION 9)

To handle the deep kernel case, we will need some more assumptions on the form of the kernel.

(II) $\phi_\omega(x) = \phi_\omega^{(\Lambda)}$ is a feedforward neural network with Λ layers given by

$$\phi_\omega^{(0)}(x) = x \quad \phi_\omega^{(\ell)}(x) = \sigma^{(\ell)}\left(W_\omega^{(\ell)} \phi_\omega^{(\ell-1)}(x) + b_\omega^{(\ell)}\right),$$

where the network parameter ω consists of all the weight matrices $W_\omega^{(\ell)}$ and biases $b_\omega^{(\ell)}$, and the activation functions $\sigma^{(\ell)}$ are each 1-Lipschitz, $\|\sigma^{(\ell)}(x) - \sigma^{(\ell)}(y)\| \leq \|x - y\|$, with $\sigma^{(\ell)}(0) = 0$ so that $\|\sigma^{(\ell)}(x)\| \leq \|x\|$. Define a Banach space on ω , with addition and scalar multiplication componentwise, and

$$\|\omega\| = \max_{\ell \in \{1, \dots, \Lambda\}} \max\left(\|W_\omega^{(\ell)}\|, \|b_\omega^{(\ell)}\|\right),$$

where the matrix norm denotes operator norm $\|W\| = \sup_x \|Wx\|/\|x\|$. (For convolutional networks, see Remark 25.)

(III) k_ω is a kernel of the form (1),

$$k_\omega(x, y) = [(1 - \epsilon)\kappa(\phi_\omega(x), \phi_\omega(y)) + \epsilon]q(x, y),$$

with $0 \leq \epsilon \leq 1$, κ a kernel function, and $q(x, y)$ a kernel with $\sup_x q(x, x) \leq Q$.

Note that this includes kernels of the form $k_\omega(x, y) = \kappa(\phi_\omega(x), \phi_\omega(y))$: take $\epsilon = 0$ and $q(x, y) = 1$.

(IV) κ in Assumption (III) is a kernel function satisfying

$$|\kappa(a, b) - \kappa(a', b')| \leq L_\kappa (\|a - a'\| + \|b - b'\|).$$

This holds for a Gaussian κ via Lemma 20.

We now turn to proving Assumption (C) for deep kernels. First, we will need some smoothness properties of the network ϕ .

Lemma 22. Under Assumption (II), suppose ω, ω' have $\|\omega\| \leq R, \|\omega'\| \leq R$, with $R \neq 1$. Then, for any x ,

$$\|\phi_\omega(x)\| \leq R^\Lambda \|x\| + \frac{R}{R-1}(R^\Lambda - 1) \quad (13)$$

$$\|\phi_\omega(x) - \phi_{\omega'}(x)\| \leq \left(\Lambda R^{\Lambda-1} \left(\|x\| + \frac{R}{R-1} \right) - \frac{R^\Lambda - 1}{(R-1)^2} \right) \|\omega - \omega'\|. \quad (14)$$

If $R \geq 2$, we furthermore have

$$\|\phi_\omega(x)\| \leq R^\Lambda (\|x\| + 2) \quad (15)$$

$$\|\phi_\omega(x) - \phi_{\omega'}(x)\| \leq \Lambda R^{\Lambda-1} (\|x\| + 2) \|\omega - \omega'\|. \quad (16)$$

The proof, by recursion, is given in Appendix A.5. We are now ready to prove Assumption (C) for deep kernels.

Proposition 23. Make Assumptions (I) to (IV) and Assumption (B), with $R_\Omega \geq 2$.⁸ Then Assumption (C) holds: for any $x, y \in \mathcal{X}$ and $\omega, \omega' \in \Omega$,

$$|k_\omega(x, y) - k_{\omega'}(x, y)| \leq 2Q(1 - \epsilon)L_\kappa \Lambda R_\Omega^{\Lambda-1}(R_X + 2)\|\omega - \omega'\|.$$

Proof.

$$\begin{aligned} |k_\omega(x, y) - k_{\omega'}(x, y)| &= (1 - \epsilon)|\kappa(\phi_\omega(x), \phi_\omega(y)) - \kappa(\phi_{\omega'}(x), \phi_{\omega'}(y))|q(x, y) \\ &\leq Q(1 - \epsilon)L_\kappa (|\phi_\omega(x) - \phi_{\omega'}(x)| + |\phi_\omega(y) - \phi_{\omega'}(y)|) \\ &\leq Q(1 - \epsilon)L_\kappa \Lambda R_\Omega^{\Lambda-1} (\|x\| + \|y\| + 4) \|\omega - \omega'\| \\ &\leq Q(1 - \epsilon)L_\kappa \Lambda R_\Omega^{\Lambda-1} (2R_X + 4) \|\omega - \omega'\|. \end{aligned} \quad \square$$

Remark 24. For the deep kernels we use in the paper (Assumptions (II) to (IV)) on bounded domains (Assumption (I)), we know L_κ via Proposition 23; Theorem 6 combines Theorem 11, Corollary 12, and Proposition 23. If we further use a Gaussian kernel q of bandwidth σ_ϕ , the last bracketed term in the error bound of Theorem 11 becomes

$$\frac{2(1 - \epsilon)}{\sigma_\phi \sqrt{e}} \Lambda R_\Omega^{\Lambda-1} (R_X + 2) + \sqrt{2 \log \frac{2}{\delta} + 2D \log(4R_\Omega \sqrt{n})}.$$

The component $R_\Omega^{\Lambda-1}(R_X + 2)$, from (15), is approximately the largest that ϕ_ω could make its outputs' norms; σ_ϕ will generally be on a comparable scale to the norm of the actual outputs of the network, so their ratio is something like the “unused capacity” of the network to blow up its inputs. This term is weighted about equally in the convergence bound with the square root of the total number of parameters in the network.

Remark 25. We can handle convolutional networks as follows. We define Ω in essentially the same way, letting $W_\omega^{(\ell)}$ denote the convolutional kernel (the set of parameters being optimized), but define $\|\omega\|$ in terms of the operator norm of the linear transform corresponding to the convolution operator. This is given in terms of the operator norm of various discrete Fourier transforms of the kernel matrix by Lemma 2 of Bibi et al. (2019); see also Theorem 6 of Sedghi et al. (2019). The number of parameters D is then the actual number of parameters optimized in gradient descent, but the radius R_Ω is computed differently.

⁸Of course, if we know a bound of $R_\Omega < 2$, the result will still hold using $R_\Omega = 2$. It is also possible to show a tighter result, via (13) and (14) or their analogue for $R = 1$; the expression is simply less compact.